

Convexity, end to end

An indexed summary of the tail-hedge investigation — the barbell, the curveball, funded vs paid convexity, carry, the diversified tail hedge, correlation structure, and the growth-vs-ruin frontier. Eleven sketches, their verdicts, the durable laws they establish, and the one idea that earned its way to a live test.

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RESEARCH THREAD — THOUGHT EXERCISES, NOT DEPLOYED

Every result below is a research sketch on ETF proxies, not a strategy managing capital, with one stated exception (multi-horizon trend, now in a paper A/B). All are reproducible under [scripts/research/](#). Read the *shapes and directions*, not the point estimates — tail-strategy backtests are dominated by which crises fell in the window, and this one holds only a single sharp crash (COVID) and a single inflation bear (2022).

SUMMARY IN ONE PARAGRAPH

We asked whether convex tail protection could be added to the Spine cheaply. It cannot — not for free. The investigation established a clean map: **convexity is either *free* (funded by the momentum premium, via trend — it protects *sustained* crises and improves return) or *paid* (funded by decay, via long-vol/short-carry — it protects *sharp* crises but bleeds, and cannot be timed)**. Along the way: a paid overlay on the Spine *hurts* it; making the trend sleeve multi-horizon is a clean win and the one idea we graduated; carry is a real but negative-skew premium that damages convex sleeves; and the best tail hedge is *diversified across crisis regimes* (deflation vs inflation), which are anti-correlated. The single durable lesson: you don't *time* the tail or *optimize* it — you budget it, diversify what it insures against, and let the free convexity in trend do the rest.

1 The index

Eleven sketches, in the order they were run. Each is a self-contained script under `scripts/research/` .

#	SKETCH	THE QUESTION	THE VERDICT
1	barbell_sketch	Does a Taleb barbell (90% safe / 10% convex) work, and its reversal — the "curveball" (90% convex)?	Barbell = paid insurance : bleeds ~3%/yr, skew +2.1, positive in every crisis (+19% COVID). Curveball = a dark horse: ruin if compounded (−44%/yr) but +237% in COVID on its 10% — hold it, let it bleed, never time it.
2	spine_tail_overlay	Does a vol-aware VIXY tail overlay improve the Spine?	No — it hurts. The Spine's trend sleeve is already a crisis hedge; VIXY carry is a drawdown machine in calm; "buy when vol is cheap" over-carries the bleed. Paid convexity on a book that already has free convexity is pure cost.
3	convex_trend	Can the trend sleeve be made more convex — for free?	Yes — multi-horizon (3/6/12mo). Sharpe 1.04→1.16, drawdown −8.9→−7.0%, more convex, no theta. The obvious "convex response function" (strength ²) backfires — it levers calm uptrends and flattens skew. Convexity must be in the payoff, not the sizing.
4	multi_horizon_split	Do multi-horizon trend and split-universe breadth stack?	Substitutes, not complements. On Sharpe they don't add (both ~1.1); the combination only wins on tail-shape. Multi-horizon on the plain 6-asset universe is the Sharpe winner and the simplest — the drop-in upgrade.
5	crash_and_curveball	Can a faster/break trend catch a sharp crash? Can the curveball be rescued?	No, and no. Faster trend fights reversal and whipsaws the V-recovery (COVID still ~0). Every rule that trims the curveball's bleed also removes its payoff (vol-gating: +237%→0%). Establishes the free-vs-paid capstone .
6	carry_sleeve	Does adding carry as a third premium diversify (the "Holy Grail")?	Uncorrelated but doesn't pay. Carry's P&L is genuinely uncorrelated to base (−0.02) / trend (−0.18), but it lost money (−2%/yr in this window), has negative skew (−2.5), and lowers the blend's Sharpe (1.16→0.96). Diversification is necessary, not sufficient.

#	SKETCH	THE QUESTION	THE VERDICT
7	carry_curveball	Can carry's income <i>fund</i> the curveball's bleed (carry as the barbell's earning sleeve)?	No — it eats the payoff. Carry crashes in the same disasters the curveball pays (COVID: carry -10% vs bills ~0), shrinking the net (+17%→+6%); in a slow bear both lose. The safe sleeve must be <i>truly</i> safe — bills doing nothing is the feature.
8	inverse_carry	What about <i>shorting</i> carry — short high-carry instead of long?	A real second hedge. Inverting flips skew -2.1→+2.1; it catches the rate/credit crisis (2022 +5%) that VIXY misses. Caveat: its positive return here is a 2022 window artifact, it's milder than VIXY, and +0.43 correlated. Tail hedges can be diversified across disaster <i>types</i> .
9	tail_hedge_diversification	Can we hedge every crisis type with one budgeted sliver?	Yes — hedge the two regimes. Deflation hedges (VIXY, short-credit) and inflation hedges (short-duration, long-commodities) are anti-correlated (-0.3), so a blended sliver is positive in <i>every</i> crisis, at a lower bleed than any single hedge — but breadth dilutes explosiveness (COVID +3% vs VIXY +19%).
10	pairs_meanreversion	Is the cross-industry correlation structure tradeable — pairs / relative-value mean-reversion?	No — the coin-flip wall. A diversified pairs book nets ~0.15 Sharpe after costs (noise); "correlated" pairs frequently drift apart permanently (KO/PEP "half-life" 3,800d). Correlation is real for <i>risk</i> , not exploitable as <i>alpha</i> — lead-lag is priced instantly (-0.10).
11	growth_vs_ruin	How aggressive can we get — Kelly leverage, and a max-aggressive bet's terminal wealth?	Aggression needs edge we lack. At 6.5% financing, leverage <i>lowers</i> growth (Kelly ≈ 1×); at 3% Kelly wants ~8× but -67% drawdown in a calm decade = a ruin trap. A max-aggressive convex bet's median → \$0.52/yr (negative growth) with a fat right tail. See §4.

2 The durable laws

Six conclusions survive the whole thread. They are directional and structural — the kind of finding robust to the small-sample caveats, unlike any single return number.

I. Convexity is free or paid — never both cheap and fast.

Free convexity (trend / time-series momentum, funded by the momentum premium) protects *sustained* crises like 2022 and *improves* return. **Paid convexity** (long-vol, short-carry, funded by decay) protects *sharp* crises like COVID but *bleeds*. No instrument tested delivers cheap, well-timed, sharp-crash protection — that is the shape of the problem, not a search we abandoned.

II. Timing the tail removes the tail.

Every rule that reduced a paid hedge's bleed also removed its crash payoff, because the bleed and the convexity are the same thing (being positioned in the decaying instrument at all times). Vol-gating the curveball took COVID from +237% to 0%. You budget paid convexity as an un-timed premium — you do not optimize *when* to hold it.

III. Diversification is necessary, but not sufficient.

An uncorrelated return stream only helps the ensemble if it is *positive-expectancy*. Carry was genuinely uncorrelated (−0.02 to base) yet lowered the blended Sharpe, because its own expectancy (in this proxy/window) was negative and its skew was the wrong sign. Low correlation is a filter, not a guarantee.

IV. A hedge's safe sleeve must be inert in a crisis.

Replacing the barbell's bills with a carry sleeve failed precisely because carry *does something* in a crisis — it crashes — eating the convex payoff. Bills doing nothing in a disaster is not a weakness; it is the entire point. Anything with hidden crisis-beta (credit, duration, "reach for yield") is not a safe sleeve, however high its calm-market yield.

V. There are two crisis regimes, and they need opposite hedges.

Deflationary crashes (equities down, bonds up, vol spikes) want VIXY and short-credit; inflationary crashes (equities *and* bonds down, like 2022) want short-duration and long-commodities. The two hedge-clusters are *anti-correlated* (-0.3), so a small sliver spread across both has no crisis blind spot and a lower net premium than any single hedge. Diversify the tail hedge across *what it insures against*, not just how much of it you hold.

VI. Aggression multiplies edge — with no edge it multiplies only ruin.

Kelly caps the useful leverage: past the optimum, growth *falls* while ruin keeps climbing, so maximum aggression is doubly irrational. For a ~ 1.0 -Sharpe strategy at 6.5% financing the optimum is $\approx 1\times$ (leverage *lowers* growth); at 3% Kelly wants $\sim 8\times$, a ruin trap (-67% drawdown in a calm decade). A max-aggressive convex bet's median terminal wealth sits *below* the stake — a lottery, not a strategy. Rational aggression is fractional Kelly on real edge, or a small barbell sleeve.

3 The two-regime map (Law V, in numbers)

Crisis capture by hedge, showing the regime split — each single hedge has a blind spot; the blend does not.

HEDGE	REGIME IT COVERS	COVID '20 (DEFLATION)	2022 (INFLATION)	CALM BLEED
VIXY (long vol)	deflationary vol spike	+19%	+1%	−7%/yr
short-credit (short HYG)	deflationary credit crisis	+18%	+18%	−8%/yr
short-duration (short TLT)	inflationary rate shock	−9%	+24%	−2%/yr
long-commodities (DBC)	inflation	−14%	+13%	+5%/yr*
diversified sliver	both regimes	+3%	+14%	−3%/yr

* long-commodities' positive carry is a 2016–2026 window artifact (the inflation era); over a normal window it is commodity beta, not a hedge.

4 Aggression & leverage — the growth-vs-ruin frontier

The mirror of the conservative question. "Most aggressive" has a principled ceiling (Kelly) and a brutal catch: aggression is a *multiplier on edge*, and ours is modest. Two pictures, from the Spine's real returns and a bootstrap Monte Carlo.

Leverage vs growth (Spine). At retail financing the optimum is to *de-lever*; at cheap financing Kelly wants ~8×, which is a drawdown trap even in a calm decade.

LEVERAGE	CAGR @3%	MAXDD @3%	CAGR @6.5%	MAXDD @6.5%
1.0×	+6.1%	−9%	+6.1%	−9%
3.0×	+11.4%	−31%	+3.9%	−38%
8.0×	+18.8%	−67%	−7.0%	−84%

Kelly-optimal (in-sample): ~8× @3% financing (a ruin trap) · ≈ 1× or below @6.5%. The −67% drawdown at 8× is in a decade with *no* catastrophe — a real 2008-tail there is a wipeout.

Terminal wealth after 1 year (5,000 bootstrapped paths, start \$1): the conservative book compounds steadily; leverage at retail cost *lowers* the median while widening the downside; the max-aggressive convex bet has a median *below* the stake with a fat right tail — a lottery.

STRATEGY	MEDIAN	MEAN	P5	P95	>2×
Spine (1×	\$1.06	\$1.06	\$0.97	\$1.17	0%
Spine (3× @6.5%)	\$1.04	\$1.05	\$0.80	\$1.38	0%
Max-aggressive (VIXY 100%)	\$0.52	\$0.73	\$0.22	\$1.80	4%

Growth and ruin rise together only up to Kelly; beyond it growth falls while ruin climbs. With modest edge and retail financing the growth-optimal leverage is ≈ 1× — **you cannot aggress your way around a coin flip**. The only rational aggression is a small, positive-skew convex sleeve inside a barbell (Law VI); the SPCX round-trip from \$225 to \$114 is the live illustration of what leveraged convex longs do when the reversal comes.

5 What graduated, and what was declined

Graduated to a live test

Multi-horizon trend (3/6/12-month agreement) — the one strict improvement (higher Sharpe, shallower drawdown, more convex, no cost, a standard untuned technique). Wired into `spine.jl` behind `BB_TREND_MODE` and now running a paper A/B on a dedicated third account (`single-sign` vs `split-sign` vs `single-multi`), pending OOS/purged-CV before any funded path.

Reviewed, understood, declined

Paid tail overlays on the Spine (redundant with its free convexity); the curveball as anything but an inert-funded, un-timed insurance sliver; the "convex response function" (backfires); carry as a return sleeve (uncorrelated but negative-expectancy) and as a barbell base (crashes when the hedge pays). Each is on record with its reason, so none needs re-litigating.

Standing, if a sharp-crash hedge is ever wanted

A small, budgeted, *diversified* tail sliver — spread across the two crisis regimes (VIXY + short-credit for deflation; short-duration + long-commodities for inflation) — held always, never timed, sized as insurance you have decided in advance you can lose. That is the honest form of the "dark horse."

6 Meta-caveats

Three limits apply to every number above and should temper any reading of the returns (the *directions* are far more trustworthy than the magnitudes):

1. One of each crisis. The 2016–2026 window contains a single sharp crash (COVID) and a single inflation bear (2022). Tail results are dominated by these two events; a different decade would move every crisis-capture and bleed figure.

2. ETF proxies, not the real instruments. Carry is proxied by income yield (which conflates it with credit/duration beta); the long-vol hedge is VIX futures (VIXY), not options; commodity carry is muddled by collateral yield. A true cross-asset version needs futures/rates data outside our reach.

3. Window-artifact returns. Several "positive" results (inverse carry, short-duration, long-commodities) earned only because 2022 fell in the sample. They are shorting or tilting against premia that are positive in the long run; do not extrapolate their carry.

Reproducible sketches (all under `scripts/research/`): `barbell_sketch.py`, `spine_tail_overlay_sketch.py`, `convex_trend_sketch.py`, `multi_horizon_split_sketch.py`, `crash_and_curveball_sketch.py`, `carry_sleeve_sketch.py`, `carry_curveball_sketch.py`, `inverse_carry_sketch.py`, `tail_hedge_diversification_sketch.py`, `pairs_meanreversion_sketch.py`, `growth_vs_ruin_sketch.py`.
Companion notes: `docs/barbell_note.pdf`, `docs/research_frontier.html`. Validated strategy: `docs/whitepaper.html` / `FINANCIAL_METHODS.md`. Decisions: `docs/CANONICAL_ARCHITECTURE.md`.

Research summary. Describes thought exercises, not strategies managing capital; not investment advice, an offer, or a solicitation. All figures are backtested/simulated on ETF proxies and are not indicative of future results; tail-strategy backtests are especially sample-dependent. Trading carries substantial risk, including loss of principal. © 2026 Carter Warrens · MIT-licensed.